

TrES-5b

R-0844

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_171009 · created 2026-07-28T19:40:45+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458035.70664 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1798 +/- 0.0097
Transit depth (Rp/Rs)^2	3.23 +/- 0.35 [%]
Orbital Inclination (inc)	89.49 +/- 0.94
Airmass coefficient 1 (a1)	1.331 +/- 0.033
Airmass coefficient 2 (a2)	-0.226 +/- 0.021
Scatter in the residuals of the lightcurve fit is	2.5 %
Best Comparison Star	None
Optimal Aperture	3.27
Optimal Annulus	8.51
Transit Duration (day)	0.09 +/- 0.001

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1798 $\pm$ 0.0097 vs 0.142 (deviation 2.82 σ)
Duration fit vs published	0.09 d vs 0.0758 d (deviation 10.24 σ)
Mid-time O-C	5.8 min
Depth SNR	13.4
Residual scatter	2.5 %

Light curve

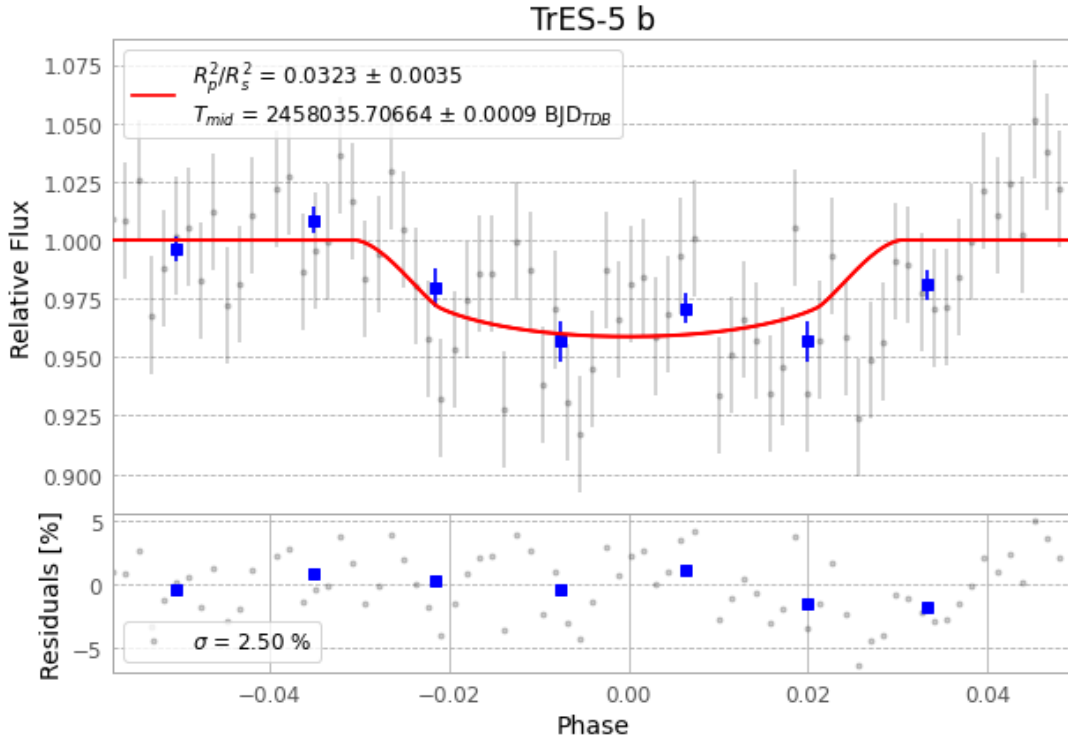


Figure 1 — FinalLightCurve_TrES-5 b_2017-10-08.png (SHA-256 04cd7af2a32c7e8e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 294], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6388e3f1cbbea05a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), TRES-5171009025112.FITS → TRES-5171009063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-171009.log (8787654e29b4...), FinalLightCurve_TrES-5 b_2017-10-08.png (04cd7af2a32c...), FinalLightCurve_TrES-5 b_2017-10-08.csv (5cc655777db2...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 19:40Z.